Question #1 of 23

Question ID: 1472207

The technique in which a machine learns to model a set of output data from a given set of inputs is *best* described as:

A) supervised learning.

B) deep learning.

C) unsupervised learning.

Joyce Tan manages a medium-sized investment fund at Marina Bay Advisors that specializes in international large cap equities. Over the four years that she has been portfolio manager, Tan has been invested in approximately 40 stocks at a time.

Tan has used a number of methodologies to select investment opportunities from the universe of investable stocks. In some cases, Tan uses quantitative measures such as accounting ratios to find the most promising investment candidates. In other cases, her team of analysts suggest investments based on qualitative factors and various investment hypotheses.

Tan begins to wonder if her team could leverage financial technology to make better decisions. Specifically, she has read about various machine learning techniques to extract useful information from large financial datasets, in order to uncover new sources of alpha.

Question #2 - 5 of 23

Question ID: 1472223

Tan is interested in using a supervised learning algorithm to analyze stocks. This task is *least* likely to be a classification problem if the target variable is:

A) continuous.

B) ordinal.

C) categorical.

Question #3 - 5 of 23

Question ID: 1472224

After Tan implements a particular new supervised machine learning algorithm, she begins to suspect that the holdout samples she is using are reducing the training set size too much. As a result, she begins to make use of K-fold cross-validation. In the K-fold cross-validation technique, after Tan shuffles the data randomly it is most likely that:

A) k – 1 samples will be used as validation samples.

B) k – 1 samples will be used as training samples.

C) the data will be divided into k – 1 equal sub-samples.

Question #4 - 5 of 23

Question ID: 1472225

At first Tan bases her stock picks on the results of a single machine-learning model, but then begins to wonder if she should instead be using the predictions of a group of models. Compared to a single machine-learning model, an ensemble machine learning algorithm is *most* likely to produce predictions that are:

A) more accurate and more stable.

B) more precise but less dependable.

C) less reliable but more steady.

Question #5 - 5 of 23

Question ID: 1472226

Tan is interested in applying neural networks, deep learning nets, and reinforcement learning to her investment process. Regarding these techniques, which of the following statements is *most* accurate?

A) Neural networks with one or more hidden layers would be considered deep learning nets (DLNs).

B) Reinforcement learning algorithms achieve maximum performance when they stay as far away from their constraints as possible.

C) Neural networks work well in the presence of non-linearities and complex interactions among variables.

Question #6 of 23

Question ID: 1508647

Considering the various supervised machine learning algorithms, a penalized regression where the penalty term is the sum of the absolute values of the regression coefficients best describes:

A) support vector machine (SVM).

B) least absolute shrinkage and selection operator (LASSO).

C) k-nearest neighbor (KNN).

Question #7 of 23

Question ID: 1508649

The unsupervised machine learning algorithm that reduces highly correlated features into fewer uncorrelated composite variables by transforming the feature covariance matrix *best* describes:

A) k-means clustering.

B) hierarchical clustering.

C) principal components analysis.

Question #8 of 23

Question ID: 1472212

Overfitting is *least likely* to result in:

A) higher number of features included in the data set.

B) higher forecasting accuracy in out-of-sample data.

C) inclusion of noise in the model.

Question #9 of 23

Question ID: 1472210

Which supervised learning model is *most appropriate* (1) when the Y-variable is continuous and (2) when the Y-variable is categorical

	Continuous Y-variable	Categorical Y-variable
A)	Decision trees	Regression
B)	Regression	Classification
C)	Classification	Neural Networks

Question #10 of 23

Question ID: 1472221

An algorithm that involves an agent that performs actions that will maximize its rewards over time, taking into consideration the constraints of its environment, *best* describes:

A) deep learning nets.

B) neural networks.

C) reinforcement learning.

Question #11 of 23

Question ID: 1472218

What is the appropriate remedy in the presence of excessive number of features in a data set?

A) Big data analysis.

B) Dimension reduction.

C) Unsupervised learning.

Question #12 of 23

Question ID: 1472208

Which of the following statements about supervised learning is *most accurate*?

A) Supervised learning requires human intervention in machine learning process.

B) Typical data analytics tasks for supervised learning include classification and prediction.

C) Supervised learning does not differentiate between tag and features.

Question #13 of 23

Question ID: 1472219

Dimension reduction is *most likely* to be an example of:

A) clustering.

B) supervised learning.

C) unsupervised learning.

Question #14 of 23

Question ID: 1472214

In machine learning, out-of-sample error equals:

A) Standard error plus data error plus prediction error.

B) bias error plus variance error plus base error.

C) forecast error plus expected error plus regression error.

Hanna Kowalski is a senior fixed-income portfolio analyst at Czarnaskala BP. Kowalski supervises Lena Nowak, who is a junior analyst.

Over the past several years, Kowalski has become aware that investment firms are increasingly using technology to improve their investment decision making. Kowalski has become particularly interested in machine learning techniques and how they might be applied to investment management applications.

Kowalski has read a number of articles about machine learning in various journals for financial analysts. However, she has only a minimal knowledge of how she might source appropriate model inputs, interpret model outputs, and translate those outputs into investment actions.

Kowalski and Nowak meet to discuss plans for incorporating machine learning into their investment model. Kowalski asks Nowak to research machine learning and report back on the types of investment problems that machine learning can address, how the algorithms work, and what the various terminology means.

After spending a few hours researching the topic, Nowak makes a number of statements to Kowalski on the topics of:

- Classification and regression trees (CART)
- Hierarchical clustering
- Neural networks
- Reinforcement learning (RL) algorithms

Kowalski is left to work out which of Nowak's statements are fully accurate and which are not.

Question #15 - 18 of 23

Question ID: 1472228

Nowak first tries to explain classification and regression tree (CART) to Kowalski. CART is *least* likely to be applied to predict a:

A) discrete target variable, producing a cardinal tree.

B) continuous target variable, producing a regression tree.

C) categorical target variable, producing a classification tree.

Question #16 - 18 of 23

Question ID: 1472229

Which of the following statements Nowak makes about hierarchical clustering is *most* accurate?

A) Bottom-up hierarchical clustering begins with each observation being its own cluster.

B) In divisive hierarchical clustering, the algorithm seeks out the two closest clusters.

C) Hierarchical clustering is a supervised iterative algorithm that is used to build a hierarchy of clusters.

Question #17 - 18 of 23

Question ID: 1472230

Which of the following statements Nowak makes about neural networks is *most* accurate? Neural networks:

A) have an input layer node that consists of a summation operator and an activation function.

B) are effective in tasks with non-linearities and complex interactions among variables.

C) have four types of layers: an input layer, agglomerative layers, regularization layers, and an output layer.

Question #18 - 18 of 23

Question ID: 1472231

Nowak tries to explain the reinforcement learning (RL) algorithm to Kowalski and makes a number of statements about it. The reinforcement learning (RL) algorithm involves an agent that is *most* likely to:

A) perform actions that will minimize costs over time.

B) take into consideration the constraints of its environment.

C) make use of direct labeled data and instantaneous feedback.

Question #19 of 23

Question ID: 1472215

A random forest is *least likely* to:

A) reduce signal-to-noise ratio.

B) provide a solution to overfitting problem.

C) be a classification tree.

Question #20 of 23

Question ID: 1472211

A rudimentary way to think of machine learning algorithms is that they:

A) “develop the pattern, interpret the pattern.”

B) “synthesize the pattern, review the pattern.”

C) “find the pattern, apply the pattern.”

Question #21 of 23

Question ID: 1472209

Which of the following about unsupervised learning is *most accurate*?

A) There is no labeled data.

B) Unsupervised learning has lower forecasting accuracy as compared to supervised learning.

C) Classification is an example of unsupervised learning algorithm.

Question #22 of 23

Question ID: 1472213

The degree to which a machine learning model retains its explanatory power when predicting out-of-sample is *most* commonly described as:

A) hegemony.

B) generalization.

C) predominance.

Question #23 of 23

Question ID: 1508648

Considering the various supervised machine learning algorithms, a linear classifier that seeks the optimal hyperplane and is typically used for classification, best describes:

A) classification and regression tree (CART).

B) support vector machine (SVM).

C) k-nearest neighbor (KNN).